RENAISSANCE PHILANTHROPY · AI FOR MATH FELLOWSHIP

Robo / Scribble

A game for learning Lean 4 formal mathematics — exploring a universe of formalosophers with a cute smart-elf.

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THE MOTIVATION

Formalizing the exercises of lower-year undergraduate maths

1

The goal

Teach the foundations a student needs to formalize the exercises from lower-year undergraduate maths courses — one guided planet per topic.

2

The tactics

Each level introduces the Lean tactics the proof calls for — `rw`, `simp`, `induction`, `funext`, `fun_prop`, `grind`.

3

The API

Introducing the related Mathlib definitions and lemmas for each topic.

Robo / Scribble turns that foundation into a game — real curriculum, learned by doing.

THE APPROACH

Learn Lean by exploring a universe, one planet at a time

Planets = topics

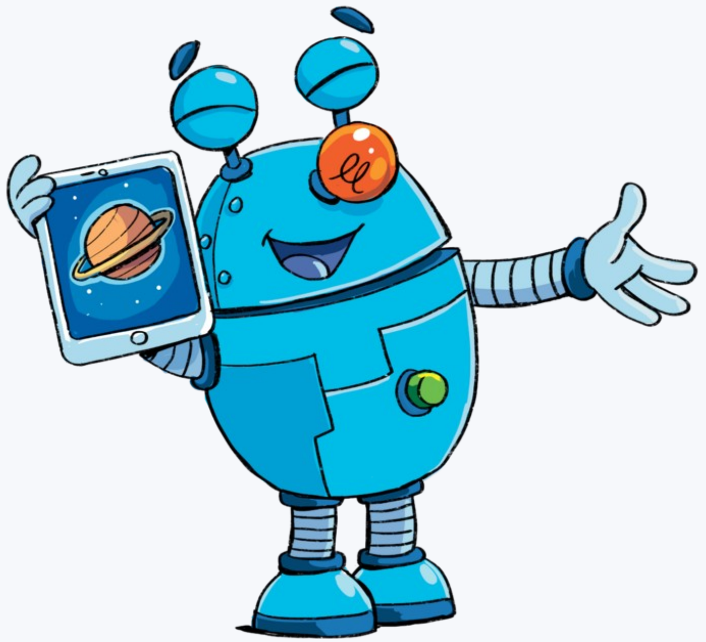
Each planet is one undergraduate topic. Levels build from a motivating idea up to a full formal proof — a boss level.

A smart-elf guide

Robo (Scribble) walks beside the player with hints and context, so intuition and Lean syntax grow together.

Tactics, in context

New Lean tactics appear exactly when the mathematics needs them — `rw`, `induction`, `funext`, `grind`, `fun_prop` and more.



Robo, the smart-elf guide, walking you through the universe.

APPROACH · IN DEPTH

Introducing `grind` — the Cafe planet

1 A planet built to teach `grind`

Cafe (replacing the old Luna) is where players first meet Lean's `grind` tactic.

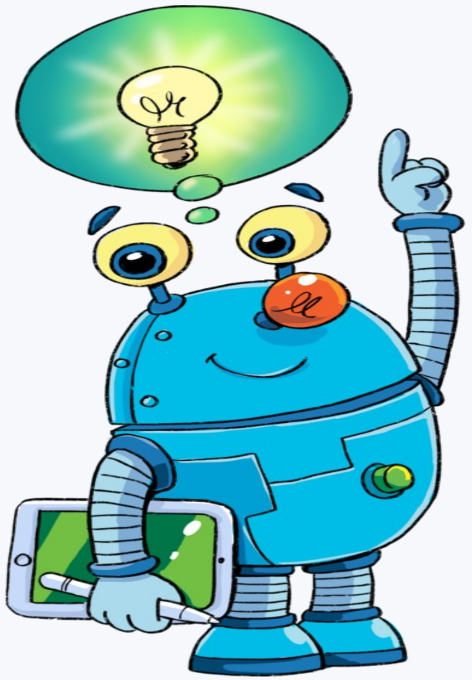
2 Avoid over-automation

`grind` is strong enough to trivialize exercises, so elsewhere levels tune it down — `ematch` := 0 and friends — keeping the reasoning with the student.

3 Closing the 'obvious' gap

`grind` dispatches statements that are clear to a mathematician but tedious to formalize by hand — a key reason we introduce it.

```
grind   -- one tactic, many inside
```



Robo, mid-idea — `grind` does the heavy lifting.

KEY RESULTS

Planets complete

12 levels shipped across three areas of the curriculum

Analysis — 6 planets

Aquarium	Bounds & suprema.
Shade	Light & shadow exercise (Intermediate value theorem).
Slope	Limits & derivatives.
Cartan	Filters and 'eventually' statements.
Fibre	No continuous $f : \mathbb{R} \to \mathbb{R}$ has every fibre of cardinality 2. (Intermediate value theorem)
Smooth	The function `exp(−1/x)`, infinitely flat at 0.

Linear Algebra — 4 planets

Hamel	Linear independence; construct an infinite independent family.
Terrace	Induction on finite set and prove that step functions are independent.
MatrixSpan	Modules and vector spaces: spanning sets.
RealUncountable	$\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$ are countable; $\mathbb{R}$ isn't finite-dim over $\mathbb{Q}$.

Quotients — 2 planets

SymmSquare	The unordered pair `Sym2` as a quotient of $A \times A$.
Quotient	The isomorphism classes of finite sets $\cong \mathbb{N}$.

PROGRESS

Where the roadmap stands

12

planets complete

across Analysis, Linear Algebra & Quotients

3

curriculum areas

each a standard topic in an undergraduate maths degree

Also in flight

Custom `grind` rolled out across the game · intervals & inequalities refactored · reviews continuing toward story.

BUILDING ON THIS WORK

Open, reusable, and ready to grow

More planets, same rails

The pipeline and framework make each new topic a repeatable build — continuity, differentiability, further quotients.

Reusable tactic patterns

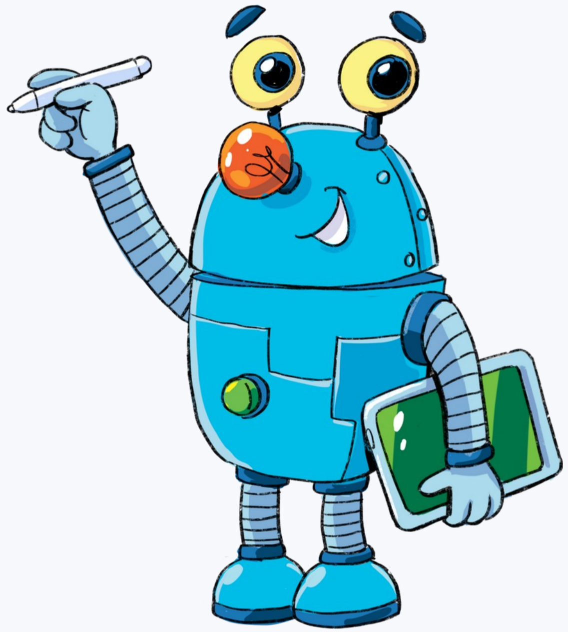
The `grind`-tuning recipe and hint-attribute conventions transfer to future levels and other Lean teaching games.

Community & translations

Fully open source (hhu-adam / Robo). New content and translations into more languages are actively welcomed.

Real classroom use

Completed planets are ready to test with students, gather feedback, and fold back in — closing the loop.



A universe still expanding.

THANK YOU

Questions?

Play the game and read the code:

`adam.math.hhu.de` `github.com/hhu-adam/Robo`